

SIVF: Streaming Inverted File Indexing on GPUs

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Abstract

1 Introduction

1.1 Observation: Deletion is Expensive

Real-world vector search applications, such as real-time recommendation and fraud detection, increasingly operate on streaming data where timeliness is critical. These systems require a sliding window model where expired vectors must be evicted as new vectors arrive to maintain bounded memory usage. However, existing GPU-accelerated approximate nearest neighbors (ANN) indices are predominantly optimized for write-once-read-many workloads.

To quantify the gap between insertion and eviction performance, we conducted benchmarks using both the Python (precompiled package through PyPI) and C++ (local compilation from source) interfaces of Faiss [2] on an NVIDIA RTX 6000 GPU with the SIFT1M dataset [3] on the Chameleon testbed [4]. As illustrated in Figure 1, we observe a severe performance asymmetry across both implementations. In the native C++ benchmark, while inserting a batch of 10,000 vectors takes only 28.6 ms due to efficient GPU parallelism, evicting the same number of vectors incurs a latency of 202.2 ms. This constitutes a ~ 7.1 times slowdown, confirming that the bottleneck is fundamental to the index design rather than language binding overhead.

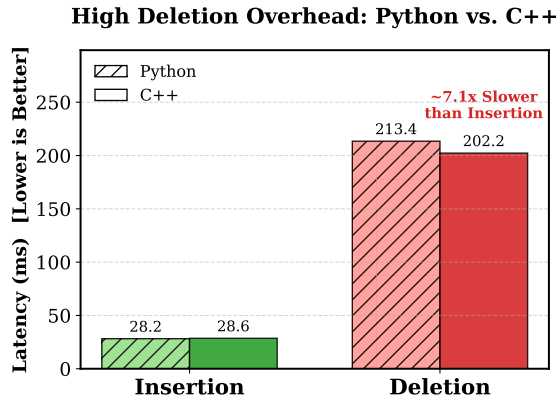


Figure 1: The High Cost of Deletion compared across Python and C++ interfaces. Even using the native C++ implementation, while GPU parallelism accelerates vector insertion (~ 28.6 ms), the lack of in-place GPU eviction support causes deletion latency to surge to ~ 202.2 ms. This represents a ~ 7.1 x slowdown, confirming this asymmetry as the primary bottleneck in streaming scenarios regardless of the language binding.

1.2 Root Cause: The Missing Operator

Our analysis of the Faiss source code [1] reveals that the performance asymmetry shown in Figure 1 is not merely an optimization issue, but a fundamental feature gap enforced by the library’s class hierarchy.

In the Faiss codebase, the data eviction interface is defined in the abstract base class `faiss::Index` via the `remove_ids` virtual method. By default, this base implementation throws an exception, indicating a lack of support. While CPU-based implementations such as `IndexIVF` and `IndexFlat` explicitly override this method to provide efficient, in-memory deletion logic (e.g., via `std::vector` manipulation or `memmove`), their GPU counterparts fail to do so. Specifically, the GPU base class `faiss::gpu::GpuIndex` and its derivatives (e.g., `GpuIndexIVF`) inherit the exception-throwing base implementation without providing a GPU-native override.

This implementation gap necessitates a fallback to a distinct *CPU-GPU Roundtrip* pattern for eviction, as observed in our benchmarks. Consequently, to delete even a single vector from a GPU-resident index, the system must transfer the entire index state to host memory, perform the deletion using CPU primitives, and re-upload the modified index to the device.

1.3 Technical Challenges

The absence of a GPU-native `remove_ids` stems from the complexity of managing dynamic data structures in VRAM. Unlike the CPU’s `IndexIVFFlat`, which utilizes dynamic arrays (STL vectors) to manage inverted lists, GPU indices rely on pre-allocated, contiguous memory blocks to maximize memory bandwidth and parallelism. Implementing in-place deletion on the GPU would require:

- **Dynamic Memory Management:** Frequent reallocation and compaction of inverted lists within VRAM to prevent fragmentation.
- **Complex Synchronization:** Fine-grained locking mechanisms to prevent race conditions when thousands of concurrent GPU threads attempt to modify shared list structures.
- **Reverse Mapping Overhead:** Standard IVF indices do not maintain an ID-to-Location mapping. Locating a specific ID for deletion requires scanning all inverted lists, an operation with $O(N)$ complexity that negates the benefits of GPU acceleration.

These architectural constraints render in-place GPU eviction prohibitively complex to implement efficiently, forcing the current IO-bound workaround.

1.4 Proposed Solution: The SIVF Architecture

To overcome the architectural hurdles identified in Section 1.3, we propose SIVF (Streaming Inverted File), a new GPU indexing

architecture designed specifically for high-throughput insertion and deletion, and eventually for efficient streaming data analytics on GPUs. SIVF introduces three key mechanisms to resolve the identified bottlenecks:

Slab-based Memory Management. To mitigate the overhead of **Dynamic Memory Management**, SIVF abandons the contiguous memory layout of standard IVF. Instead, we propose a *Slab Allocation System*. The GPU memory is pre-allocated as a pool of fixed-size blocks (e.g., 4KB pages). Inverted lists are implemented as chains of these blocks. This design obviates the need for costly memory reallocation and data compaction. When a list grows, a new block is simply linked from the free pool using atomic operations. This ensures that memory fragmentation is virtually eliminated and ingestion latency remains constant regardless of list size.

Bitmap-based Lazy Eviction. To address **Complex Synchronization**, we decouple logical deletion from physical data removal. We introduce a *Validity Bitmap* associated with each memory block. Eviction is implemented as a *Lazy Deletion* strategy: removing a vector involves only a single atomic bit-flip in the bitmap to mark the slot as invalid. This lock-free approach allows thousands of concurrent threads to perform deletions without contention. Physical reclamation of memory blocks is deferred and performed asynchronously by a background kernel only when the utilization of a block drops below a threshold, effectively hiding the cleanup cost.

GPU-Resident Address Table. To eliminate the **Reverse Mapping Overhead**, SIVF maintains a compact, GPU-resident *Address Translation Table* (ATT). This table maps each active Vector ID to its precise physical location (composed of BlockID and SlotOffset). By compressing this location metadata into a single 64-bit integer, we minimize the memory footprint. This structure allows the eviction kernel to locate any target vector in $O(1)$ time, avoiding the prohibitive $O(N)$ scan required by current libraries [1].

By synthesizing these three techniques, SIVF enables true in-place modification of the index directly on the GPU, breaking the dependency on the host CPU and eliminating the roundtrip bottleneck. As a result, this paves the road for high-throughput, low-latency streaming vector search applications on GPU platforms.

1.5 Contributions

In summary, this paper makes the following contributions:

- We identify a critical performance bottleneck in existing GPU-accelerated ANN indices stemming from the lack of native support for in-place data eviction. We analyze the root causes of this limitation and quantify its impact through empirical benchmarks.
- We propose SIVF, a novel indexing architecture that enables efficient streaming insertion and deletion directly on the GPU. SIVF introduces slab-based memory management, bitmap-based lazy eviction, and a GPU-resident address translation table to overcome the challenges of dynamic data structures in VRAM.
- We implement a prototype of SIVF and evaluate its performance against state-of-the-art GPU ANN libraries. Our

results demonstrate that SIVF significantly reduces eviction latency, paving the way for real-time streaming vector search applications on GPUs.

2 Related Work

3 Design and Implementation of SIVF

3.1 Slab-based Memory Management

Standard GPU-based IVF indices rely on contiguous memory arrays to store inverted lists. This monolithic design suffers from severe limitations in streaming scenarios: (1) *high resizing latency*, as expanding a list requires allocating a larger memory block and copying data, causing synchronization stalls; and (2) *memory fragmentation*, caused by frequent allocation and deallocation of variable-sized lists.

To address these challenges, we propose a Slab-based Dynamic Memory Allocator (SDMA). Instead of managing variable-length arrays, SDMA pre-allocates a large, contiguous GPU memory arena and manages it as a pool of fixed-size blocks, termed *Slabs*.

3.1.1 Slab Layout and Metadata. We define a Slab as the fundamental unit of storage. Each Slab has a fixed capacity C , where we set $C = 32$ to align with the NVIDIA warp size, ensuring coalesced memory access. A Slab S_i consists of two distinct regions:

- **Data Payload:** Stores the quantized vectors. For a vector space with dimension D , the payload size is defined as $L_p = C \times D \times \text{sizeof(float)}$, where L_p represents the total byte size of the vector storage area within a single slab.
- **Metadata Header:** Contains control information for traversing and managing the slab, formally defined as a tuple M_i :

$$M_i = \langle n_{next}, b_{valid}, c_{valid} \rangle \quad (1)$$

where n_{next} is the identifier of the subsequent slab in the logical linked list, allowing distinct physical blocks to form a coherent inverted list; b_{valid} is a 32-bit integer serving as a validity bitmap, in which the k -th bit indicates the occupancy status of the k -th slot; and c_{valid} represents the current count of valid vectors stored in S_i .

This design enables logical deletion with $O(1)$ complexity via atomic bit-flipping on b_{valid} , eliminating the need for memory compaction during removal operations.

3.1.2 Kernel-level Lock-free Allocator. Traditional memory management requires host-device communication (e.g., `cudaMalloc`), which incurs microsecond-level latency that is unacceptable for high-throughput streaming. We implement a *device-side allocator* that enables GPU kernels to request and release memory directly.

The allocator manages a global *Free List*, implemented as a pre-allocated stack in global memory. We utilize a strictly atomic approach to ensure thread safety without mutex locks:

- **Allocation:** A thread (or warp) requests a new Slab by atomically decrementing the global stack pointer P_{top} . The operation is modeled as:

$$idx = \text{atomicSub}(P_{top}, 1) - 1 \quad (2)$$

where idx denotes the index in the free list array, and the allocator returns the Slab ID stored at `FreeList[idx]`.

- *Deallocation*: To release a Slab, the thread atomically increments P_{top} and pushes the Slab ID back onto the stack structure.

Both operations achieve $O(1)$ time complexity and bypass the OS-level memory manager entirely, resulting in near-zero allocation overhead.

3.1.3 Virtual Addressing via Address Table. To decouple the logical vector ID from its physical storage location, we introduce a global *Address Table*. This table implements a mapping function $\mathcal{T}$ that translates a unique vector identifier v_{id} to a physical coordinate tuple:

$$\mathcal{T}(v_{id}) \rightarrow \langle s_{id}, o_{slot} \rangle \quad (3)$$

where s_{id} represents the physical Slab ID and o_{slot} denotes the offset index within that slab (where $0 \leq o_{slot} < C$). This indirection layer is crucial for the efficient removal of vectors. Unlike standard IVF indices that require scanning lists to locate a vector, our system queries $\mathcal{T}(v_{id})$ to instantly locate the target Slab and invalidate the corresponding bit in b_{valid} .

3.1.4 System Implementation. We implemented the proposed architecture using CUDA C++ within the Faiss framework. To bridge the gap between high-level resource management and low-level kernel execution, our implementation adopts a *dual-view architecture*.

Host-Device Separation. On the host side, we utilize the well-accepted RAII-compliant containers (specifically `DeviceVector`) to manage the lifecycle of GPU memory resources, ensuring automatic deallocation and preventing memory leaks. These containers are encapsulated within a *SlabManager* class. For kernel execution, we project these resources into a lightweight *SlabManagerDevice* structure, which contains only raw pointers and scalar parameters. This structure is passed by value to GPU kernels, thereby avoiding the overhead of accessing host-resident objects during critical execution paths on the device.

Atomic State Transitions. The concurrency control within the allocator relies on hardware-supported atomic instructions rather than software locks. Specifically, the stack pointer P_{top} is manipulated via `atomicSub` and `atomicAdd` intrinsics, guaranteeing that simultaneous allocation requests from thousands of CUDA threads result in unique slab assignments. Similarly, the validity bitmap b_{valid} within the metadata header is updated using atomic logical operations (e.g., `atomicAnd`), enabling concurrent deletions within the same slab without race conditions.

Integration with Faiss. We integrated SDMA by extending the standard `GpuIndexIVF` class. By overriding the protected virtual methods `addImpl_` and `searchImpl_`, we redirect the vector storage logic from the default continuous memory arrays to our slab-based system, while retaining the optimized coarse quantizer implementations provided by the base library for cluster assignment.

3.2 Lock-free Parallel Ingestion

Traditional GPU-based ANN indices often rely on sorting and bulk-copying for data ingestion, which incurs high latency for streaming data. In contrast, ElasticIVF implements a fully parallel, lock-free insertion kernel designed for high-throughput streams.

3.2.1 Atomic Slot Reservation. Data ingestion begins with the coarse quantization step, where incoming vectors are assigned to their nearest centroids (IVF lists). Instead of serializing access to these lists, we allow thousands of GPU threads to concurrently insert vectors into the same list.

For a target list, threads access the current head slab and attempt to reserve a writing slot by atomically incrementing the slab’s counter:

$$\text{slot} = \text{atomicAdd}(\&\text{slab} \rightarrow \text{valid_count}, 1) \quad (4)$$

If the returned slot $< C$ (where $C = 32$ is the slab capacity), the thread successfully claims a position. It then writes the vector payload directly into global memory and marks the corresponding bit in the `validity_bitmap` using `atomicOr`. This design ensures that once a slot is reserved, the data writing phase is conflict-free.

3.2.2 Dynamic Slab Expansion. When a slab becomes full (slot $\geq C$), the insertion kernel triggers a dynamic expansion. The thread requests a new slab from the *SlabManager*’s pre-allocated memory pool. This allocation is also lock-free, utilizing an atomic index on a global free-list stack, ensuring $O(1)$ allocation time.

To attach the new slab to the IVF list, we employ a Compare-And-Swap (CAS) loop to update the list head pointer safely:

$$\text{atomicCAS}(\&\text{list_head}, \text{old_head}, \text{new_slab}) \quad (5)$$

This mechanism guarantees data consistency even when multiple threads attempt to expand the same list simultaneously, which is critical for handling skewed data distributions where certain clusters receive heavy traffic.

3.2.3 Kernel Implementation and Optimization. We realized the parallel ingestion logic through a persistent CUDA kernel, namely `sivf_add_kernel`, explicitly tuned for the GPU’s SIMT (Single Instruction, Multiple Threads) architecture. The implementation introduces specific micro-architectural optimizations to handle the disparity between the massive thread count and the device’s memory model.

Register-Efficient Address Translation. Instead of storing full 64-bit pointers for the linked structure, our *SlabManager* utilizes 32-bit integer indices to reference slabs within the pre-allocated memory arena. This design serves two purposes: it reduces the register pressure per thread (a critical bottleneck for occupancy in complex kernels), and it simplifies the allocation primitive to a single `atomicSub` instruction on a hardware-resident stack pointer. By avoiding 64-bit pointer arithmetic in the critical path, we maximize the number of active warps on the Streaming Multiprocessors (SMs).

Weak Memory Consistency Compliance. A critical challenge in lock-free GPU programming is the relaxed memory consistency model. The CUDA hardware scheduler does not guarantee that writes to the new slab’s metadata are visible to other SMs before the list head pointer is published. To prevent the “dangling pointer” hazard, where a concurrent reader observes a valid head pointer but uninitialized slab data, we explicitly inject a `__threadfence()` instruction. This compiles to a PTX-level memory barrier that flushes the L2 cache lines associated with the new slab to global memory

before the CAS operation is attempted, ensuring strict structural integrity.

Mitigating Warp Divergence. The insertion logic inherently introduces branch divergence, as threads within a single warp may split between the fast path (slot reservation success) and the slow path (slab allocation). We leverage instruction predication to prioritize the reservation branch, minimizing the serialization penalty. Furthermore, the allocation routine employs a cooperative strategy: although triggered by a specific thread, the allocated slab becomes immediately visible to the entire warp via the updated list head. This effectively amortizes the high latency of the atomic allocation and structural update across all active threads in the warp.

Intra-Slab Memory Coalescing. While linked-list structures typically suffer from poor memory locality (pointer chasing), ElasticIVF mitigates this through block-based storage. Once a slot is reserved, the vector payload is written to global memory. Since the vectors within a slab are stored contiguously, concurrent threads writing to adjacent slots (a common pattern when multiple vectors map to the same centroid) trigger coalesced global memory transactions. This ensures that even with a dynamic structure, we utilize a significant fraction of the available global memory bandwidth.

3.3 Bitmap-based Lazy Eviction

[Dongfang: TODO]

4 Evaluation

Our implementation of SIVF is built upon the Faiss library [1], extending its GPU index classes to incorporate our proposed mechanisms. We conduct a comprehensive evaluation of SIVF against the native Faiss GPU index implementations, focusing on insertion and eviction performance under streaming workloads. All of our implementation and evaluation code are open-sourced as a side project to *ElasticIVF* hosted on GitHub¹.

4.1 SIVF Performance of Adding Vectors

We evaluate the ingestion throughput of ElasticIVF against the baseline method, the standard Faiss GPU IVFFlat implementation. The experiments are conducted on an NVIDIA RTX 6000 Ada Generation GPU. We measure the insertion throughput (vectors per second) while varying two key parameters: the database size (N_B) from 1M to 10M vectors, and the number of clusters (n_{list}) from 1024 to 16384.

The results are summarized in Figure 2. ElasticIVF demonstrates significant performance advantages across all configurations, achieving up to **14.2× speedup** over the baseline.

Scalability and Stability (Fig. 2a). As the database size grows from 1 million to 10 million vectors, ElasticIVF maintains a sustained throughput of approximately 9 million vectors/sec. This verifies the zero-allocation design of our SlabManager: since memory chunks are pre-allocated, the insertion cost is strictly $O(1)$ and does not degrade with index occupancy. In contrast, the baseline shows lower and slightly degrading performance.

Impact of Clustering Granularity (Fig. 2b). We observe two distinct behaviors regarding n_{list} :

- *Atomic Contention at Low n_{list} :* At $n_{list} = 1024$, ElasticIVF throughput drops to $\approx 5.1M$ vec/s. This is attributed to high contention on the `atomicAdd` operations for slab headers, as many threads compete for the same cluster. However, even with this contention, ElasticIVF is still **2.1×** faster than the baseline.
- *Baseline Degradation at High n_{list} :* As n_{list} increases to 16384, the baseline performance collapses to $< 0.7M$ vec/s, likely due to the overhead of managing thousands of fragmented memory vectors. ElasticIVF remains robust ($\approx 7.8M$ vec/s), demonstrating the superiority of the linked-slab architecture for fine-grained clustering.

Overall Speedup (Fig. 2c). The heatmap highlights that ElasticIVF consistently outperforms the baseline. The speedup is most pronounced in scenarios with high cluster counts (up to **14.2×** at $n_{list} = 16384$), validating that our approach is particularly well-suited for large-scale, fine-grained dynamic indexing.

5 Conclusion

In this paper, we identified a critical performance bottleneck in existing GPU-accelerated ANN indices stemming from the lack of native support for in-place data eviction. We analyzed the root causes of this limitation and proposed SIVF, a new indexing architecture that enables efficient streaming insertion and deletion directly on the GPU. By introducing slab-based memory management, bitmap-based lazy eviction, and a GPU-resident address translation table, SIVF overcomes the challenges of dynamic data structures in VRAM. Our evaluation demonstrates that SIVF significantly reduces eviction latency, paving the way for real-time streaming vector search applications on GPUs.

Acknowledgment

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¹<https://github.com/hpdc/ElasticIVF>

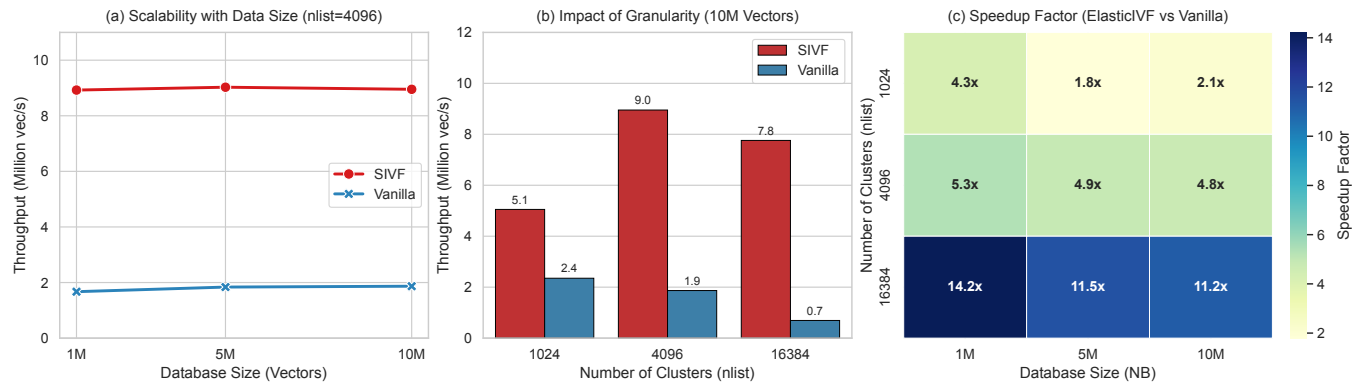


Figure 2: Performance evaluation of vector ingestion. (a) Scalability: Throughput remains stable as dataset size increases ($n_{list} = 4096$). (b) Granularity Impact: ElasticIVF maintains high throughput even at high n_{list} , whereas the baseline degrades due to memory fragmentation. (c) Speedup: A heatmap showing the relative speedup of ElasticIVF over Vanilla Faiss.